

Project Design Phase

Solution Architecture

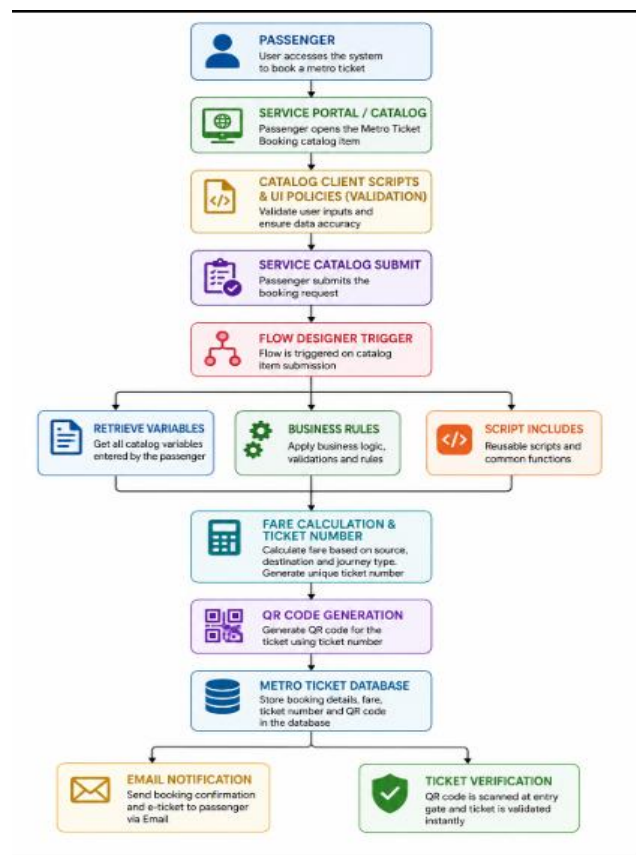
Date	25 th July 2026
Project Name	Metro Ticket Booking System

Architecture Description

The Metro Ticket Generation System follows a layered architecture that separates user interaction, business logic, data processing, and storage.

Passengers access the system through the Service Portal and submit ticket booking requests using the Service Catalog. Catalog Client Scripts and UI Policies validate user input before submission. Once submitted, Flow Designer retrieves the catalog variables and creates a booking record in the custom Metro Ticket table. Business Rules generate the ticket number, calculate fares, and create the QR code. Finally, notifications are sent to the passenger, and the ticket information is stored securely for future verification.

Solution Architecture



Components

Presentation Layer

- Service Portal
- Service Catalog
- Catalog Variables

Responsibilities

- Display booking form.
- Accept user input.
- Show booking confirmation.

Business Logic Layer

- Flow Designer
- Business Rules
- Script Includes
- Catalog Client Scripts
- UI Policies

Responsibilities

- Validate data.
- Calculate fare.
- Generate ticket number.
- Generate QR code.
- Execute booking workflow.

Data Layer

- Metro Ticket Table

- Metro Station Table
- User Table
- Booking Records

Responsibilities

- Store passenger details.
- Store booking records.
- Store ticket information.
- Maintain journey history.

Notification Layer

- Email Notifications
- Event Management
- Flow Designer Notification Actions

Responsibilities

- Send booking confirmation.
- Notify passengers.
- Send ticket details.